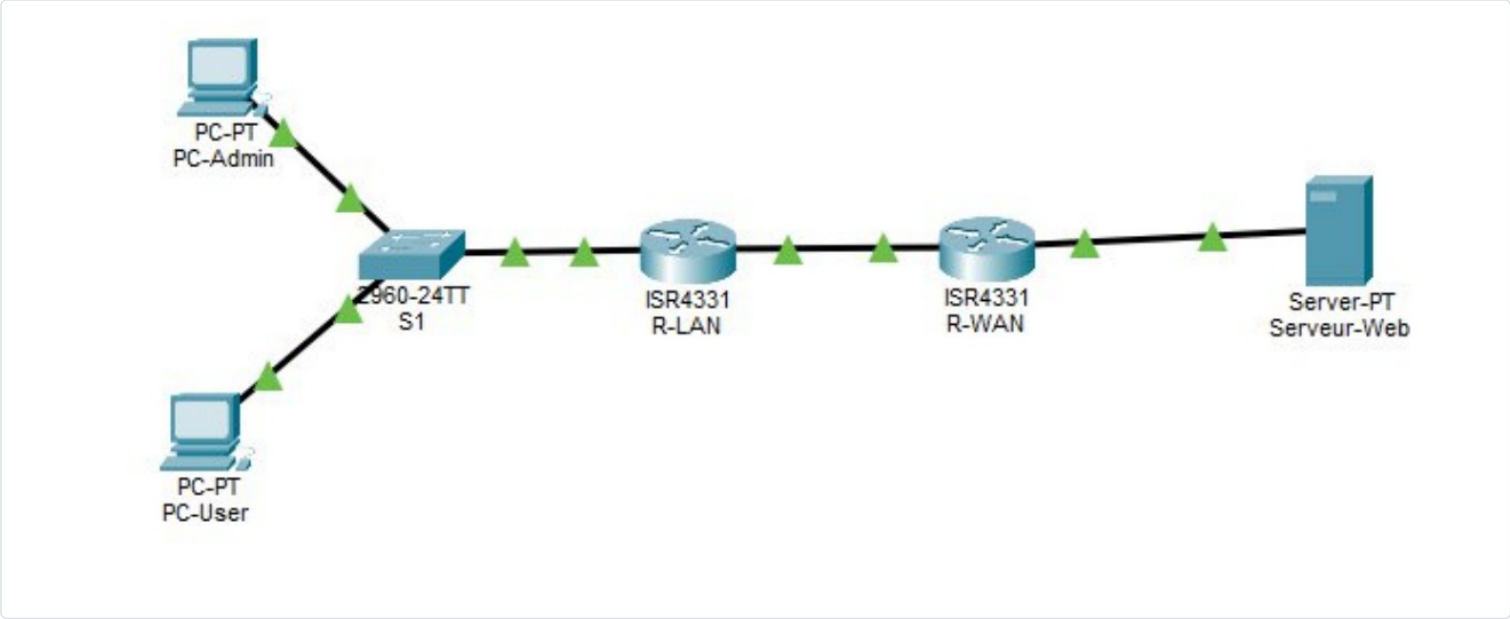


Extended ACL & NAT/PAT Perimeter Security

Scope: Cisco IOS network configuration guide **Focus:** Implementation, fallback static routing, and granular Layer 4 filtering under Cisco IOS



This guide covers the complete deployment and security methodology for a Cisco IOS network infrastructure. The goal is twofold: allow a private LAN to reach public Internet resources via dynamic NAT/PAT (Overload) translation, while enforcing surgical filtering with an **Extended Access Control List (ACL)**. This scenario closely matches the performance-based question (PBQ) requirements of the Cisco CCNA 200-301 exam.

Part 1 — Core Concepts

1. NAT/PAT (Overload) Address Translation

Under RFC 1918, private IP addresses used inside enterprise networks are not routable on the public Internet. To work around this, **NAT (Network Address Translation)** operates at the edge router. The **Overload** option (PAT — Port Address Translation) lets hundreds of local hosts share a single outbound public IP address (the router's WAN interface). The router maintains a dynamic translation table by rewriting the source TCP/UDP port of each outbound session on the fly. This conserves IPv4 address space and prevents unsolicited inbound connections from being initiated from outside.

2. Extended Access Control Lists

Unlike standard ACLs (numbered 1–99), which only filter on source IP address, **extended ACLs** (numbered 100–199) provide full granularity. They inspect Layer 3 (Network) and Layer 4 (Transport) headers to permit or deny a packet based on:

- Source host IP address
- Final destination IP address
- Specific IP protocol (IP, TCP, UDP, ICMP, OSPF, etc.)
- Destination application port (e.g. port 80 for HTTP, port 443 for HTTPS)

📌 **Golden rule:** an extended ACL should always be placed as close as possible to the traffic source (inbound on the LAN gateway interface). This design avoids wasting router CPU and bandwidth carrying packets that will ultimately be dropped further downstream.

Part 2 — Topology & Addressing

1. Physical Architecture

- **Internal LAN:** trusted internal enterprise segment
- **Inter-router WAN segment:** point-to-point interconnection simulating the link to the ISP
- **External network (Internet):** untrusted external zone hosting the legitimate target server

2. IP Addressing Matrix

Segment	Details

Enterprise network	192.168.1.0/24 — mask 255.255.255.0 — gateway 192.168.1.254 (R-LAN's Gi0/0/0)
PC-Admin	Static IP 192.168.1.10 — network administration host (access permitted)
PC-User	Static IP 192.168.1.20 — standard workstation (access restricted)
Private WAN link	10.0.0.0/30 — R-LAN (Gi0/0/1) = 10.0.0.1 R-WAN (Gi0/0/1) = 10.0.0.2
Public Internet	8.0.0.0/8 (simulated) — Web-Server = 8.8.8.8 (mask 255.0.0.0) — R-WAN gateway (Gi0/0/0) = 8.8.8.254

Part 3 — Cisco IOS Configuration Procedures

1. Full Configuration of Router R-LAN

Initializes the interfaces, applies inside/outside NAT tagging, injects the default static route, configures PAT overload, and deploys the extended filtering ACL.

```
Router> enable
Router# configure terminal
Router(config)# hostname R-LAN

! --- 1. INTERFACE CONFIGURATION & NAT TAGGING ---
R-LAN(config)# interface gigabitEthernet 0/0/0
R-LAN(config-if)# description IN_LAN_INTERFACE
R-LAN(config-if)# ip address 192.168.1.254 255.255.255.0
R-LAN(config-if)# ip nat inside
R-LAN(config-if)# no shutdown
R-LAN(config-if)# exit

R-LAN(config)# interface gigabitEthernet 0/0/1
R-LAN(config-if)# description OUT_WAN_INTERFACE
R-LAN(config-if)# ip address 10.0.0.1 255.255.255.252
R-LAN(config-if)# ip nat outside
R-LAN(config-if)# no shutdown
R-LAN(config-if)# exit

! --- 2. DEFAULT ROUTE (GATEWAY OF LAST RESORT) ---
R-LAN(config)# ip route 0.0.0.0 0.0.0.0 10.0.0.2

! --- 3. NAT OVERLOAD (PAT) CONFIGURATION ---
R-LAN(config)# access-list 1 permit 192.168.1.0 0.0.0.255
R-LAN(config)# ip nat inside source list 1 interface gigabitEthernet 0/0/1 overload

! --- 4. EXTENDED SECURITY ACL CONFIGURATION ---
R-LAN(config)# access-list 101 permit tcp host 192.168.1.10 host 8.8.8.8 eq www
R-LAN(config)# access-list 101 permit icmp host 192.168.1.10 host 8.8.8.8
R-LAN(config)# access-list 101 deny ip host 192.168.1.20 host 8.8.8.8
R-LAN(config)# access-list 101 permit ip any any

! --- 5. APPLY THE ACL TO THE INTERFACE ---
R-LAN(config)# interface gigabitEthernet 0/0/0
R-LAN(config-if)# ip access-group 101 in
R-LAN(config-if)# exit
R-LAN(config)# write memory
```

2. Provider WAN Router Configuration (R-WAN)

This router simulates the ISP's behavior. It holds a static return route back to the private LAN to allow translated packets to be forwarded correctly.

```
Router> enable
Router# configure terminal
Router(config)# hostname R-WAN

! --- 1. INTERFACE CONFIGURATION ---
R-WAN(config)# interface gigabitEthernet 0/0/1
R-WAN(config-if)# description LINK_TO_R-LAN
R-WAN(config-if)# ip address 10.0.0.2 255.255.255.252
R-WAN(config-if)# no shutdown
R-WAN(config-if)# exit

R-WAN(config)# interface gigabitEthernet 0/0/0
R-WAN(config-if)# description LINK_TO_INTERNET_SERVER
R-WAN(config-if)# ip address 8.8.8.254 255.0.0.0
R-WAN(config-if)# no shutdown
R-WAN(config-if)# exit

! --- 2. RETURN STATIC ROUTE TO THE PRIVATE LAN ---
R-WAN(config)# ip route 192.168.1.0 255.255.255.0 10.0.0.1
R-WAN(config)# write memory
```

Part 4 — Validation, Audit & Troubleshooting

1. Test Matrix & Expected Results

- **PC-Admin validation (192.168.1.10):** `ping 8.8.8.8` must succeed 100% of the time. Opening a web browser to `http://8.8.8.8` must display Cisco Packet Tracer's default landing page.
- **PC-User validation (192.168.1.20):** `ping 8.8.8.8` must immediately return an explicit `Destination host unreachable` error generated by the local R-LAN gateway. The web browser must time out.

2. Essential Verification Commands (CLI)

Run the following commands in privileged mode on R-LAN to inspect protocol behavior:

- `R-LAN# show access-lists` — displays every rule in ACL 101. Check the match counter at the end of each line — the router increments it for every intercepted packet. This is the best way to visually prove the block is effective.
- `R-LAN# show ip nat translations` — displays the active address translation table. You should see the dynamic mapping between the administrator's private IP, the assigned ephemeral TCP port, and the translated public outbound IP.
- `R-LAN# show ip nat statistics` — displays a NAT status summary (number of active translations, associated standard ACL ID, and inside/outside interface reminder).

🛡️ **Security note:** Cisco applies a sequential **first-match** filtering mechanism — line order is critical. Had `permit ip any any` been declared above the `deny ip host 192.168.1.20 ...` line, the standard user's PC would have matched the global allow rule first and bypassed the enterprise security policy entirely.